

YOUTH UNEMPLOYEMENT IN DEVELOPING COUNTRIES

An Exploratory Data Analysis

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Agendas

- Introduction
- Research Context and Objectives
- Data Description
- Data Cleaning and Prerocessing Pipeline
- Key Findings
- Policy Recommendations
- Conclusion





Background

Context

The Challenge

- 64M+ youth unemployed globally (90% in developing countries)
- Ages 15–24, actively seeking work

Why It Matters

- Economic: Reduced tax revenue, consumer spending
- Social: Crime, poverty, political instability
- Human capital: Brain drain of talented youth

Purpose

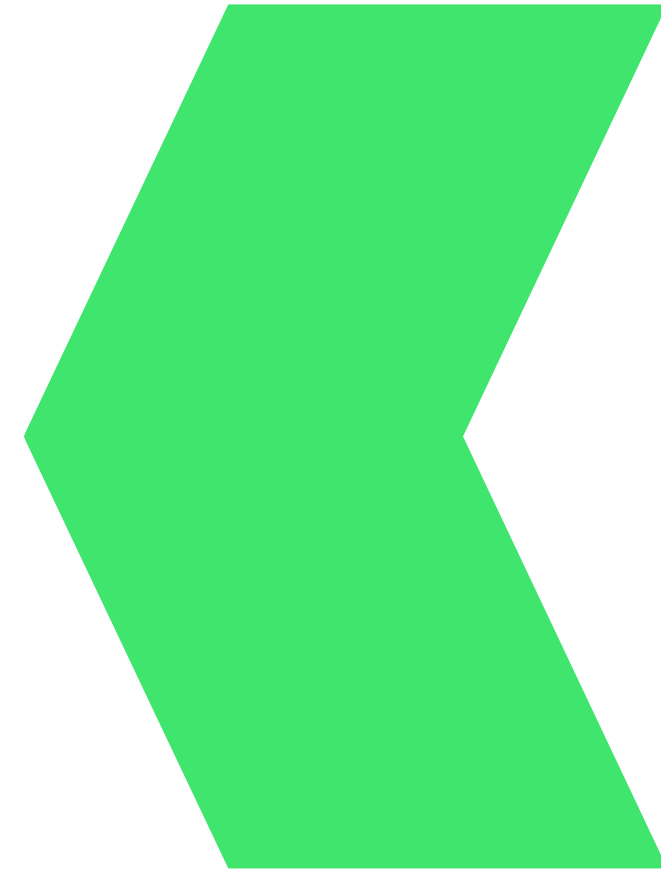
The Challenge

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Data Cleaning And Preprocessing Pipeline



01

Cleaning:

Standardized country names,
removed duplicates

- Applied mapping dictionary for name variations
- Eliminated income groups and regional aggregates
- Excluded territories and non-sovereign entities
- Verified unique country-year combinations

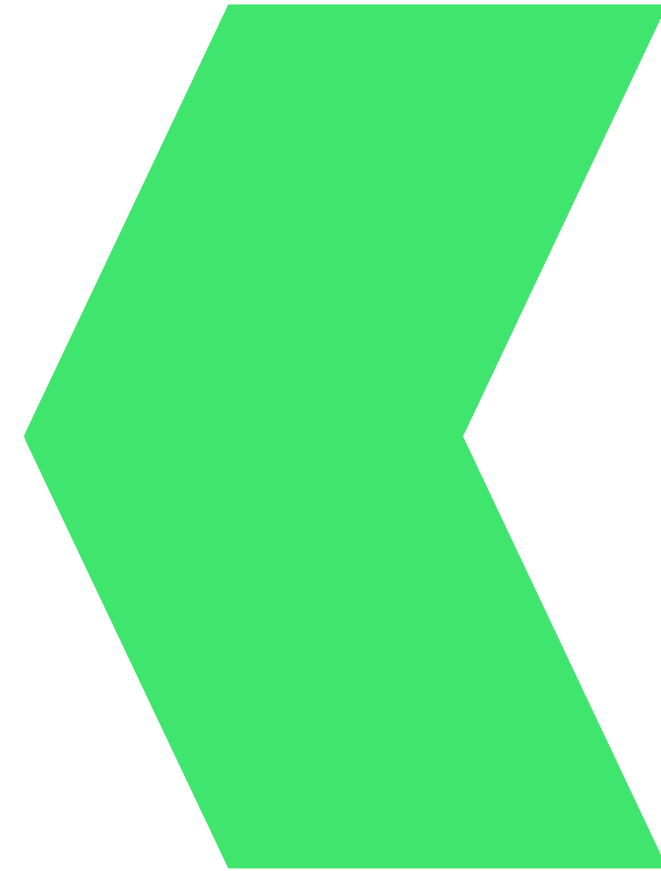
02

Missing Values:

Multi-phase imputation
(interpolation, median fill)

- Replaced placeholders ('.', 'N/A') with NaN
- Forward/backward fill within countries
- Linear interpolation for time series
- Country-specific and global median fill
- Result: 0% missing values

Data Cleaning And Preprocessing Pipeline



03

Transformation:

Z-score normalization for comparability

- Converted data types (categorical, integer, float)
- Renamed columns to snake_case format
- Standardized all numerical features
- Ensured consistency across measurement scales



04

Validation:

Quality assurance and final checks

- Verified 177 valid countries
- Confirmed 2,665 observations (2010–2022)
- Validated data integrity across 29 variables
- Ensured no duplicates or anomalies

Analysis Techniques



Distribution analysis

- Box plots
- Donut Chart
- Bar Charts(Radial. Bivariate, Grouped)

Correlation analysis

- heatmaps
- arc diagrams

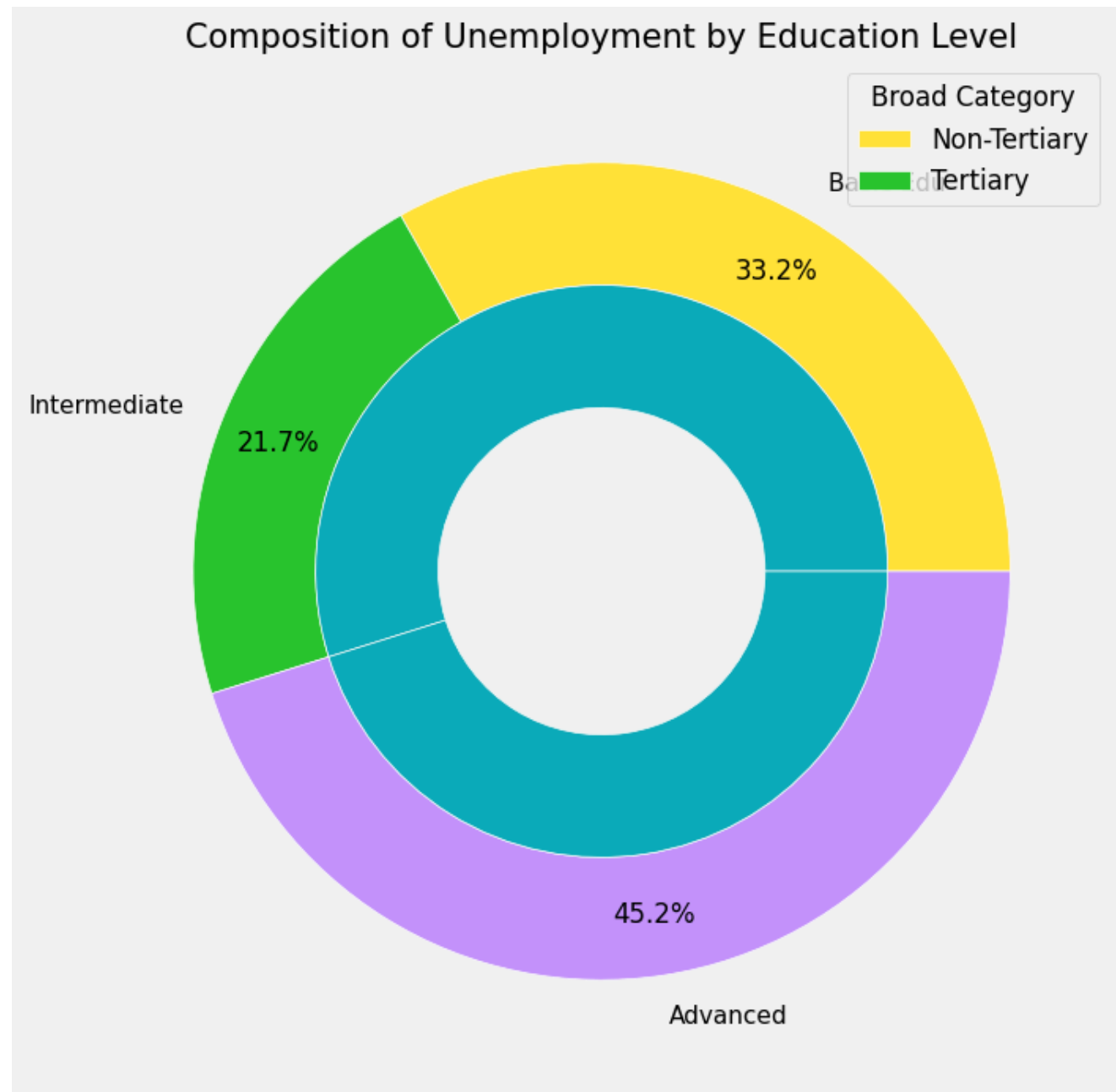
Geographic visualization

- Choropleth maps

Multivariate analysis

- 3D scatter plots
- radar charts

Composition of Unemployment



45.2%[↑]

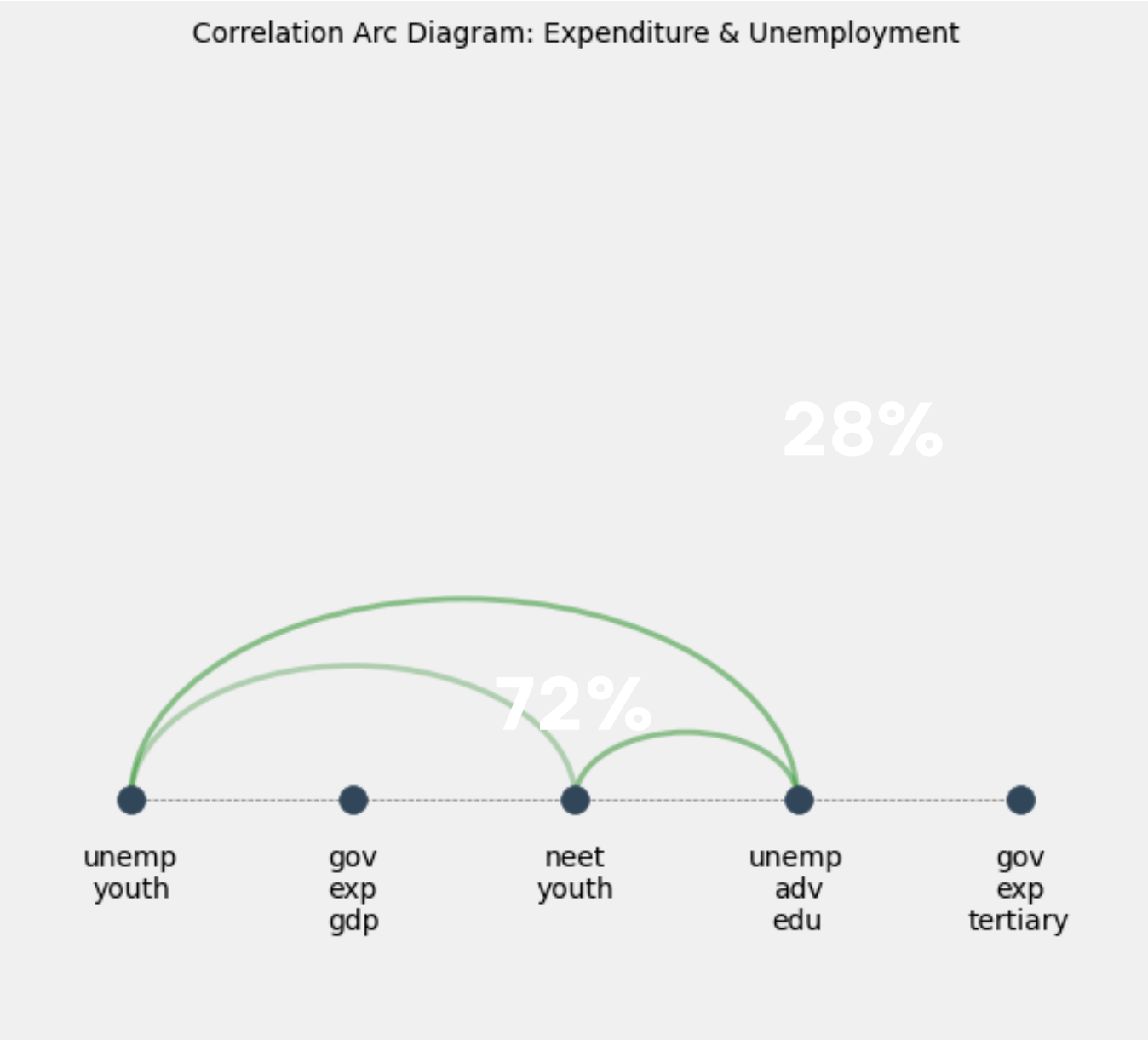
Share of Unemployed with
Advanced Education

33.2%[↑]

Share of Unemployed with
Basic Education

Arc Diagram (Correlation Analysis)

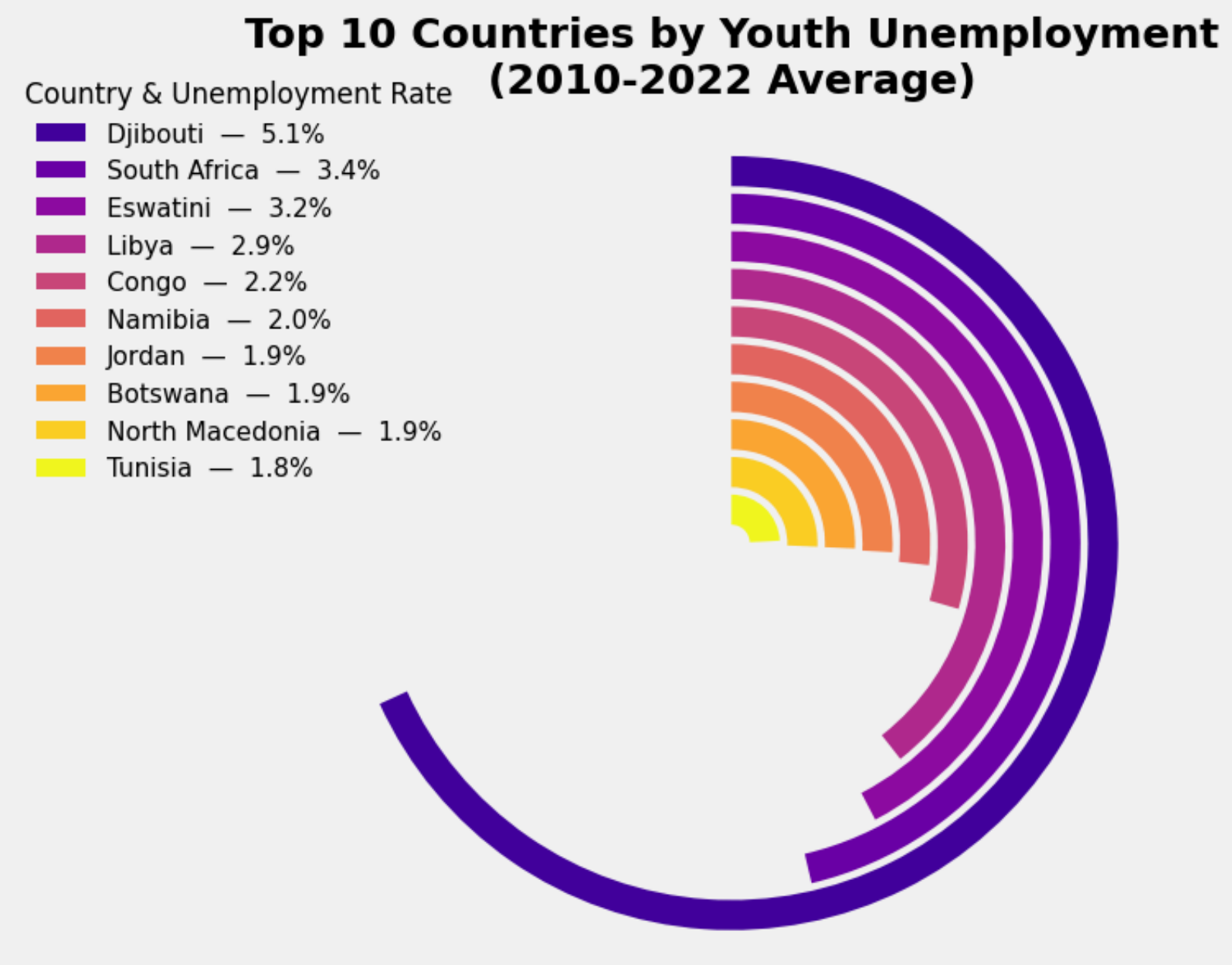
The strong connection between youth unemployment and NEET rates signals a high risk of social exclusion. As unemployment rises, youth are more likely to drop out of the system entirely rather than return to education.



The connections between Government Expenditure and employment outcomes are weak or absent in several areas. This indicates fiscal inefficiency, where funding for education is not effectively translating into job creation for graduates.



Radial Bar Chart (Top 10 Countries)

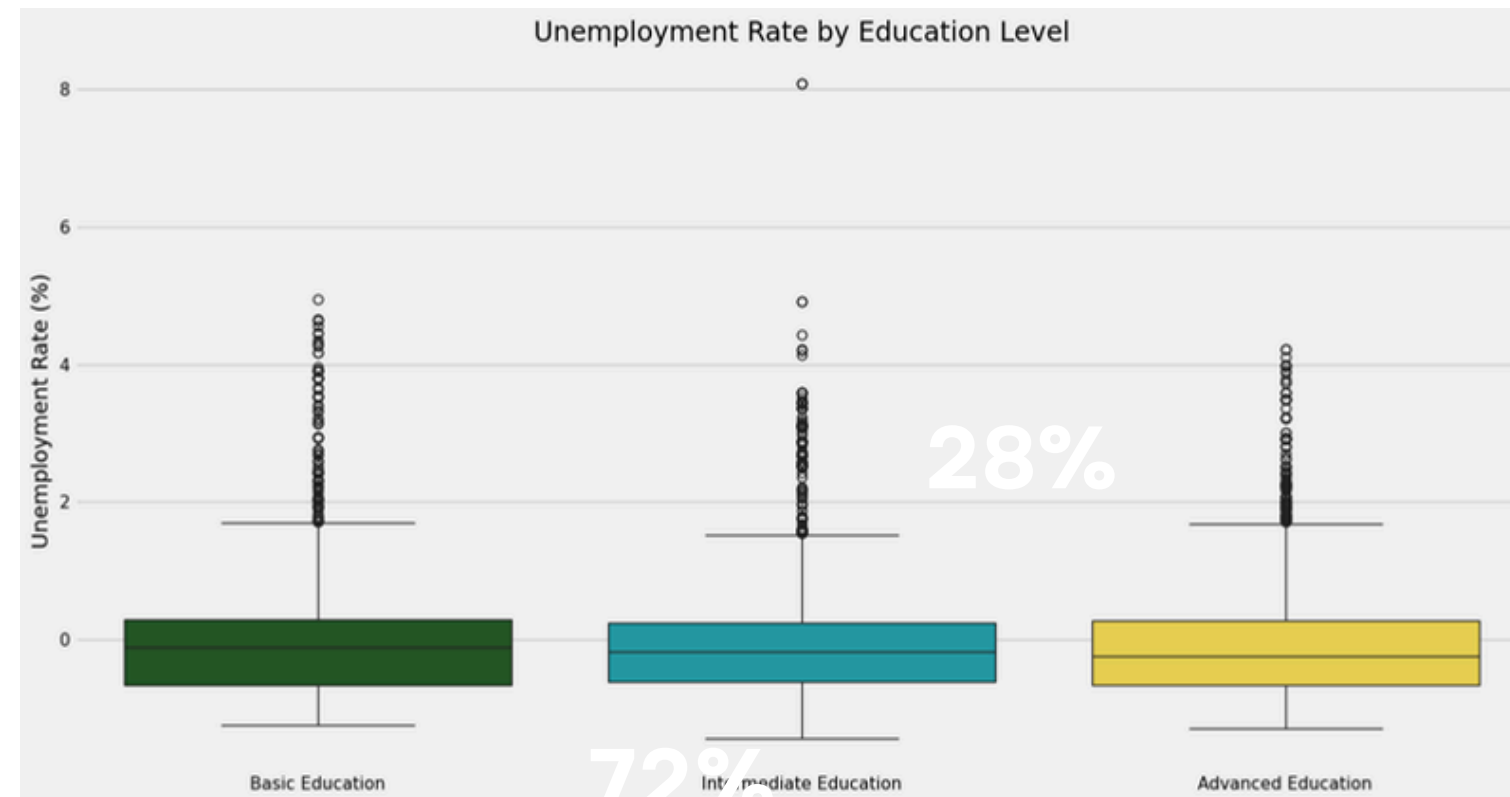


Youth unemployment is heavily concentrated in African nations (e.g., Djibouti, South Africa, Eswatini), indicating a regional crisis driven by the "youth bulge" demographic. The visual disparity between the outer rings (Djibouti) and inner rings highlights that even among the "worst performers," the severity varies significantly, requiring tailored intervention strategies.

Box Plot (Education vs. Unemployment)

The Skills Mismatch

Ideally, higher education should guarantee lower unemployment. However, the significant overlap between the "Basic" and "Advanced" boxes indicates that a degree offers limited protection against market volatility in these developing economies.



Advanced Education Outliers

The "Advanced Education" category displays alarming outliers (the circles above the top whisker). This confirms that in many nations, university graduates face exceptionally high unemployment rates due to a saturation of white-collar candidates.

Bivariate Heatmap: Govt Education Spending vs Youth Unemployment

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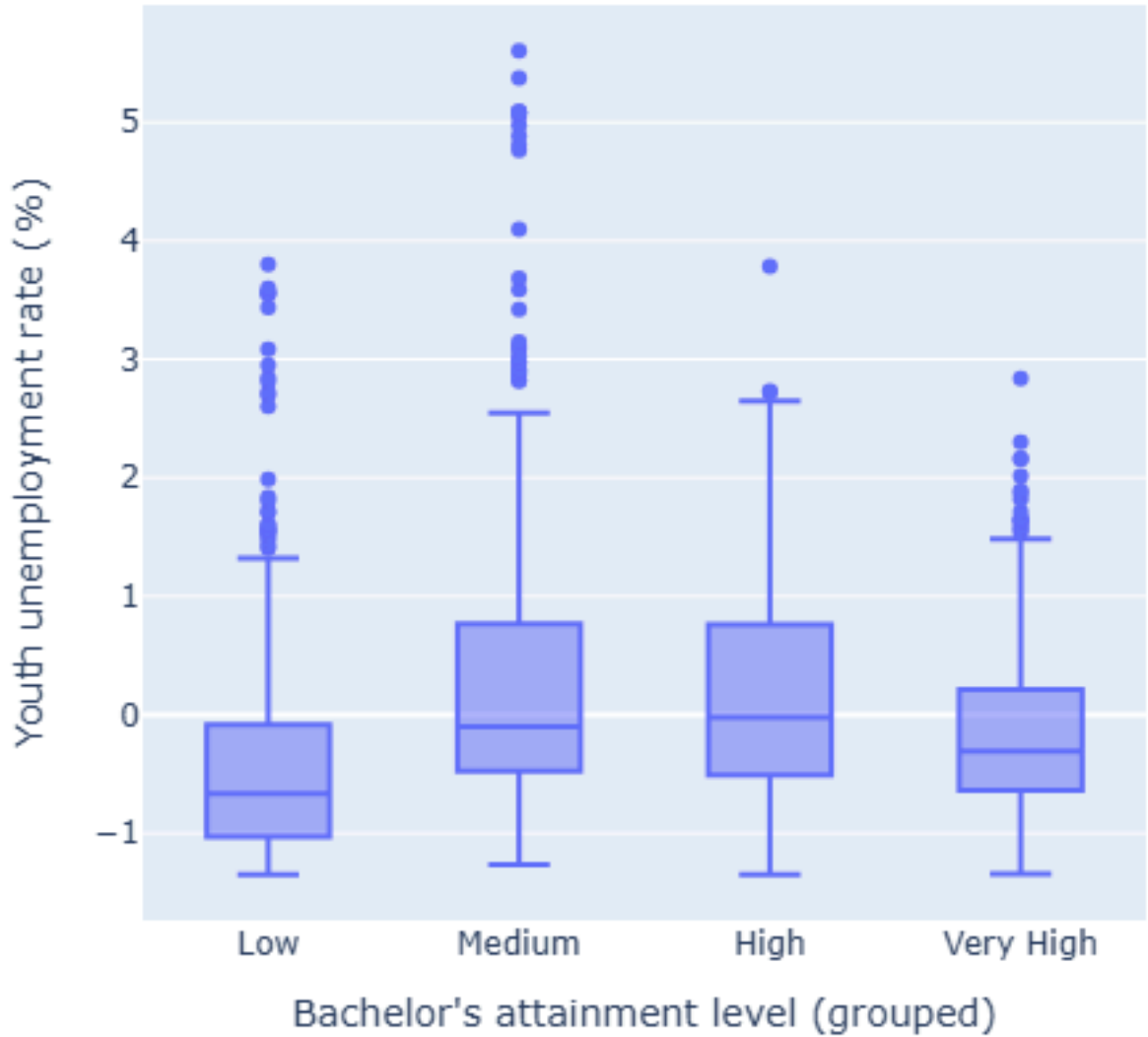


Higher education spending does not necessarily reduce youth unemployment. Many countries invest more, yet young people still face difficulty entering the labour market.

This shows that spending alone is not enough. The Job market absorption, economic structure, and skill relevance matter more than budget increases.

Bivariate Box Plot: Youth Unemployment vs Bachelor's Attainment

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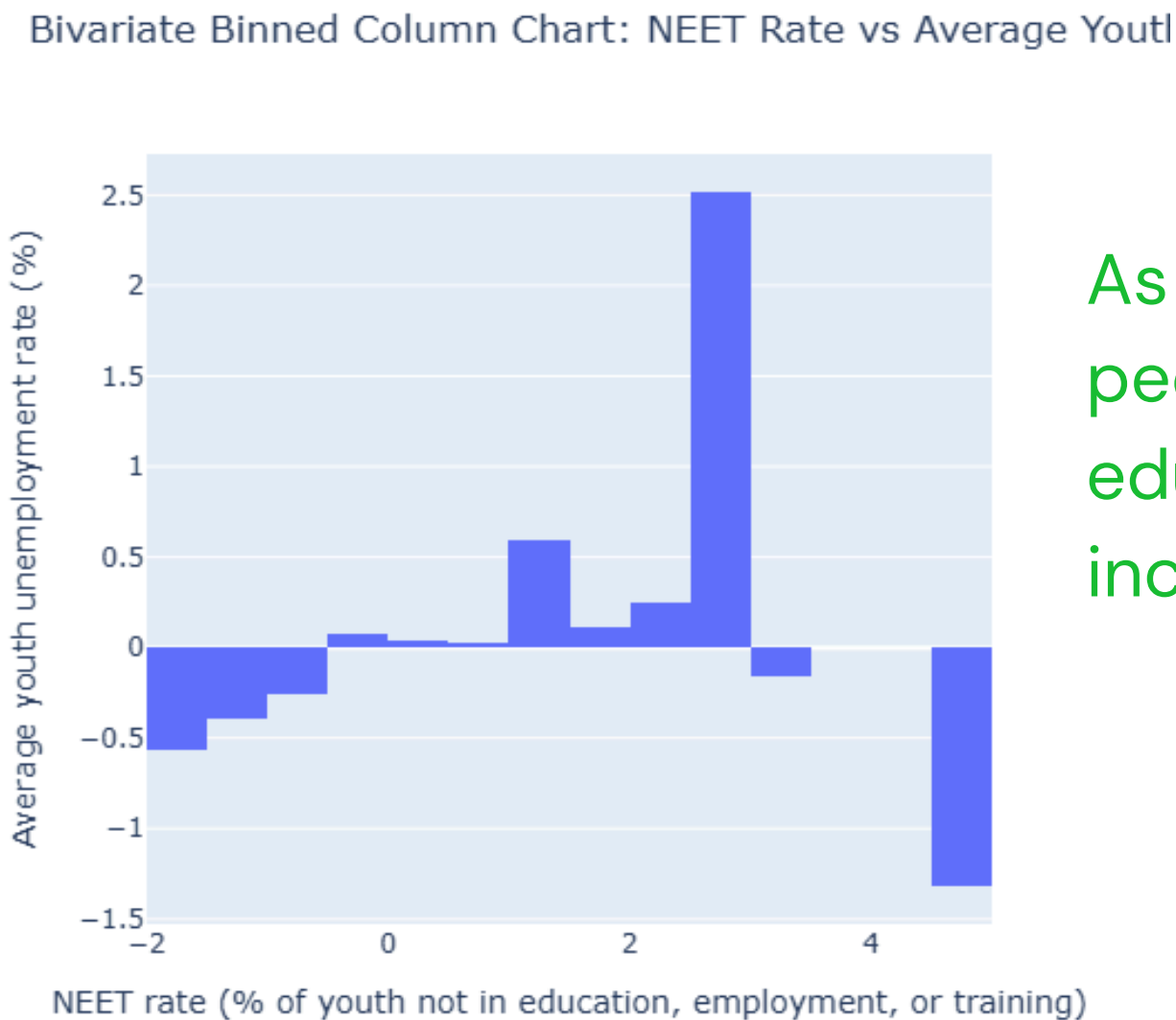


Higher educational attainment does not guarantee employment. Even countries with many graduates show high unemployment levels.

This reflects a major skills mismatch, where educational output does not match labour market demand, leading to unemployment even among degree holders.

Bivariate Binned Column Chart: NEET Rate vs Average Youth Unemployment

Countries with high NEET rates tend to have much higher youth unemployment, showing a strong positive relationship.



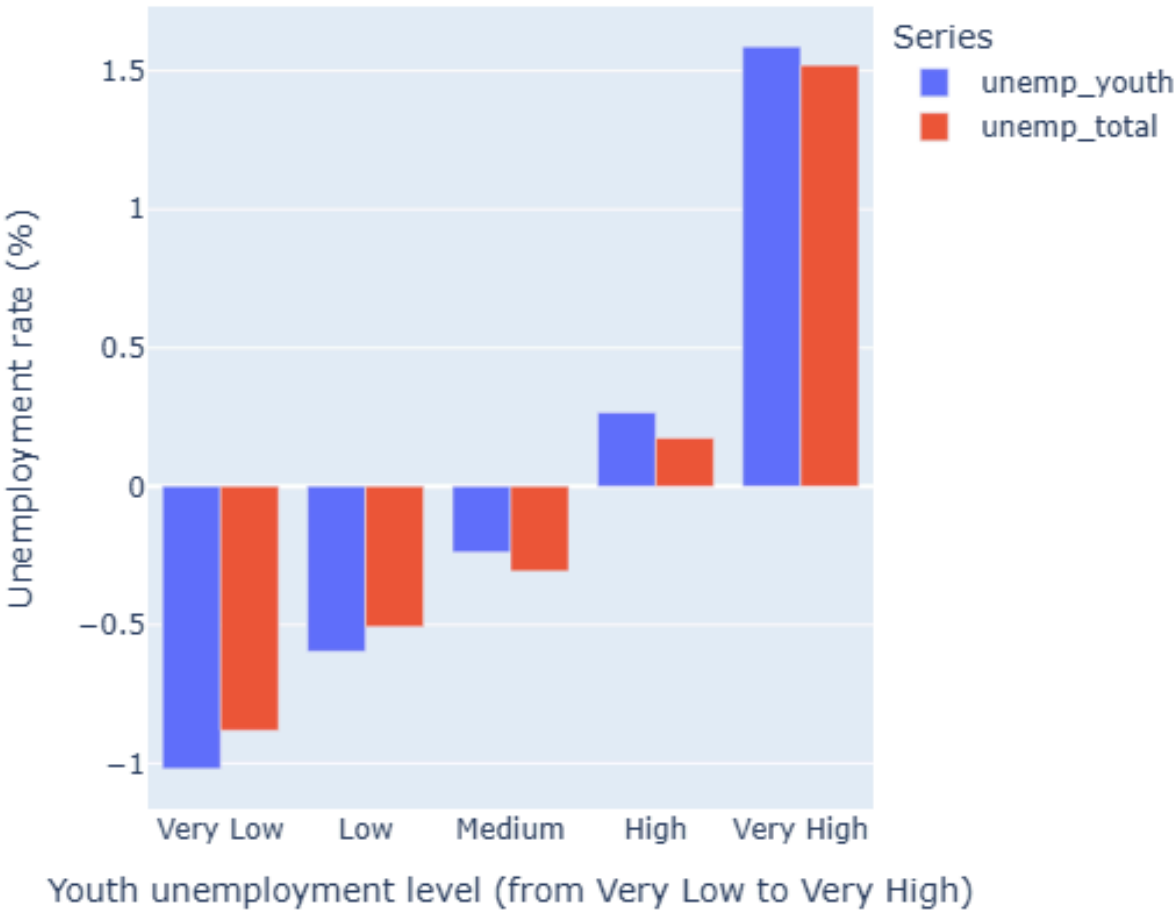
As unemployment rises, many young people fully disengage from education, training, and work, increasing long-term exclusion risks.

Grouped Bar Chart: Youth vs Total Unemployment by Youth-Unemployment Level

Youth unemployment is consistently 2–3 times higher than general unemployment across all categories.

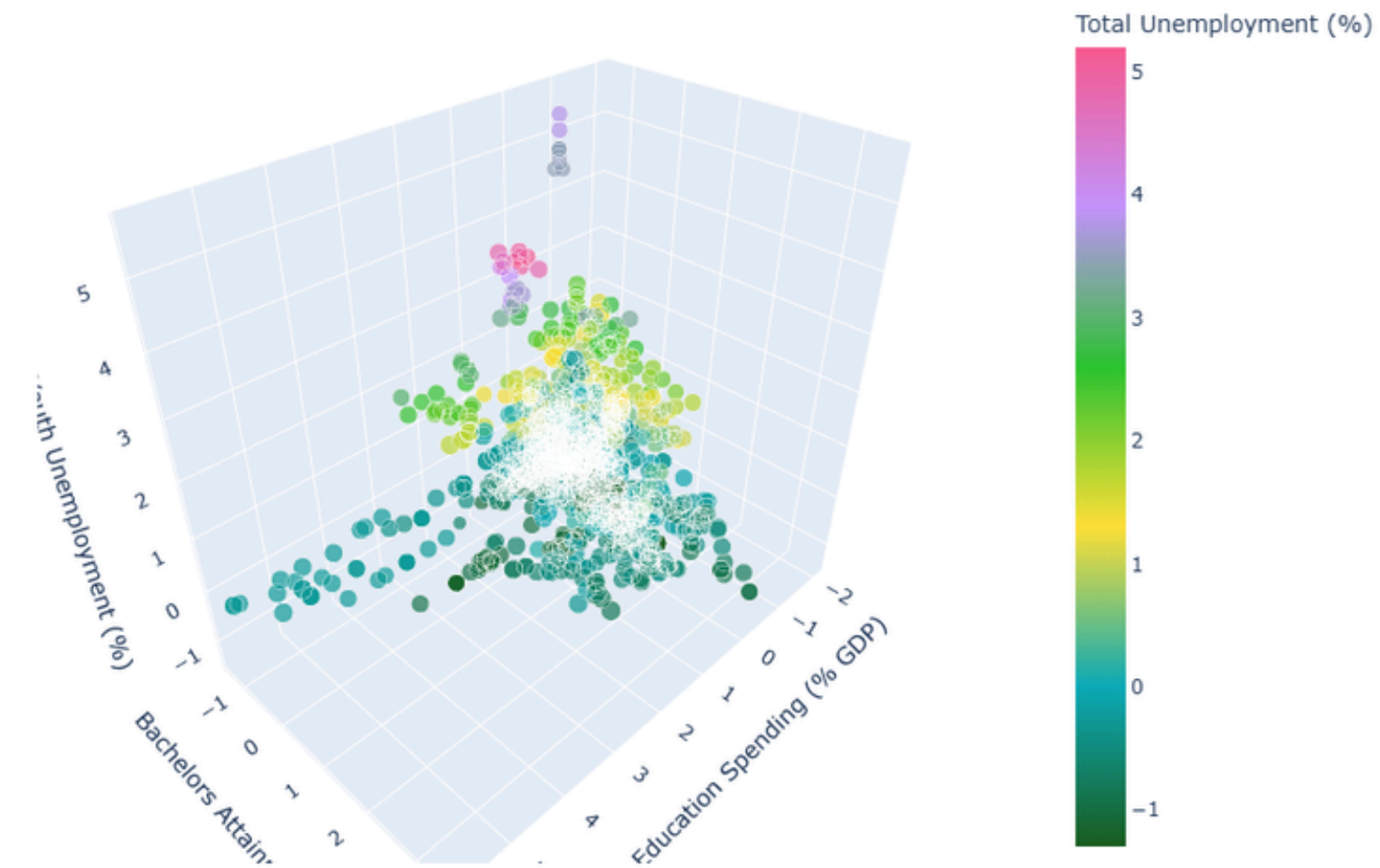
Young people face greater barriers—limited experience, slower job creation, and higher competition, making them more vulnerable than adults.

Grouped Bar Chart: Youth vs Total Unemployment by Youth-

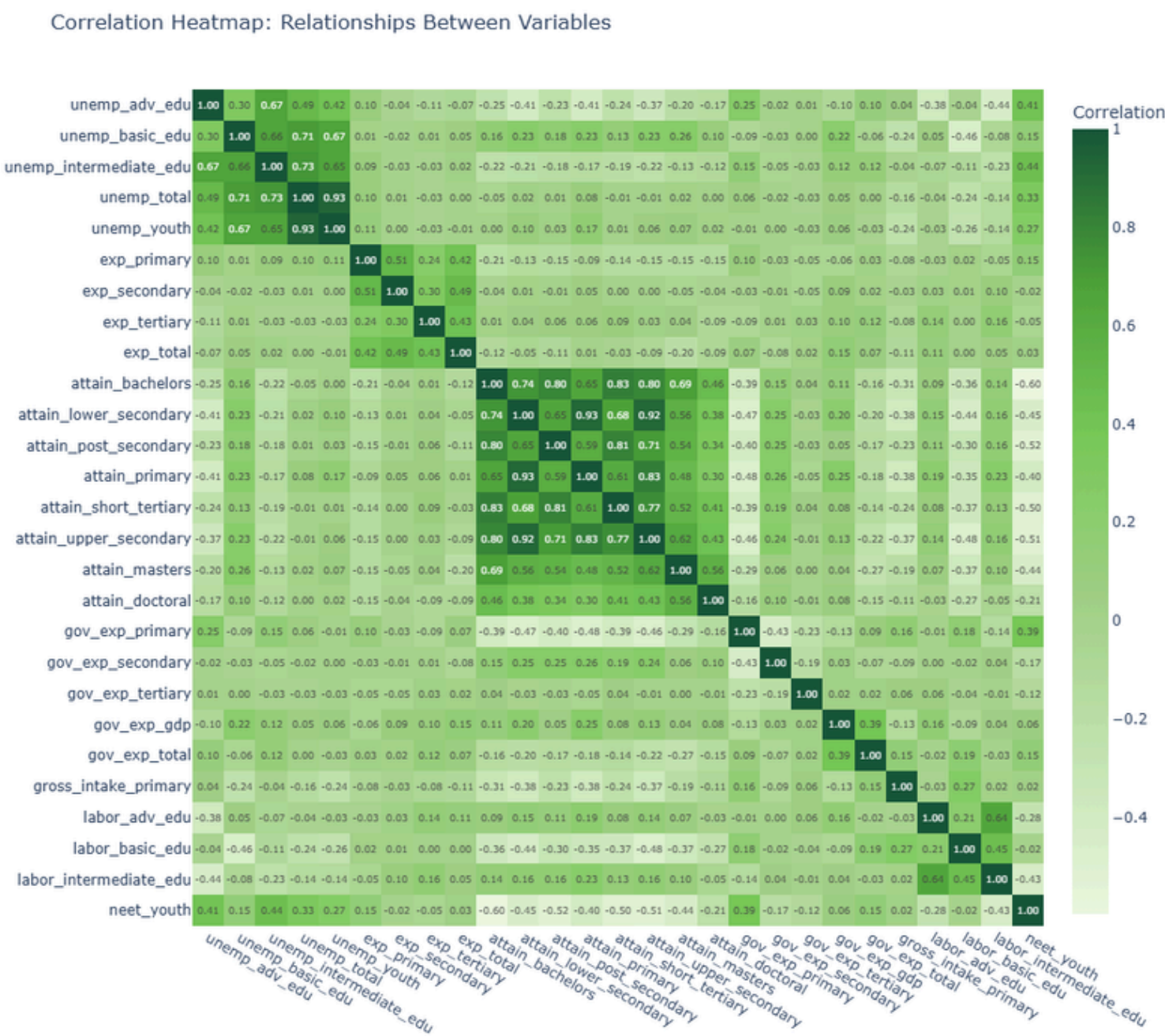


3D Scatter: Government Spending vs Education vs Youth Unemployment

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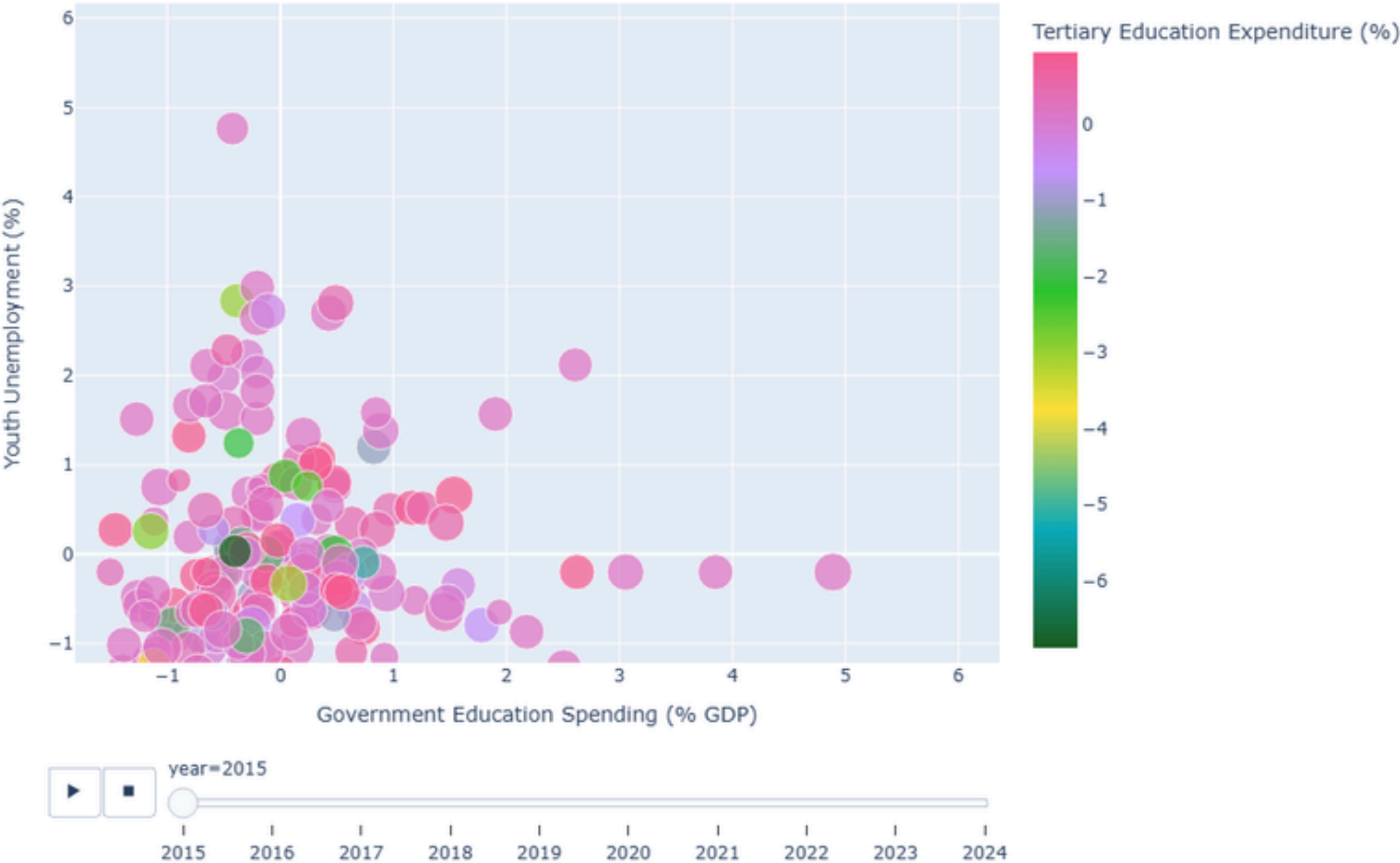


Correlation Heatmap: Relationships Between Variables

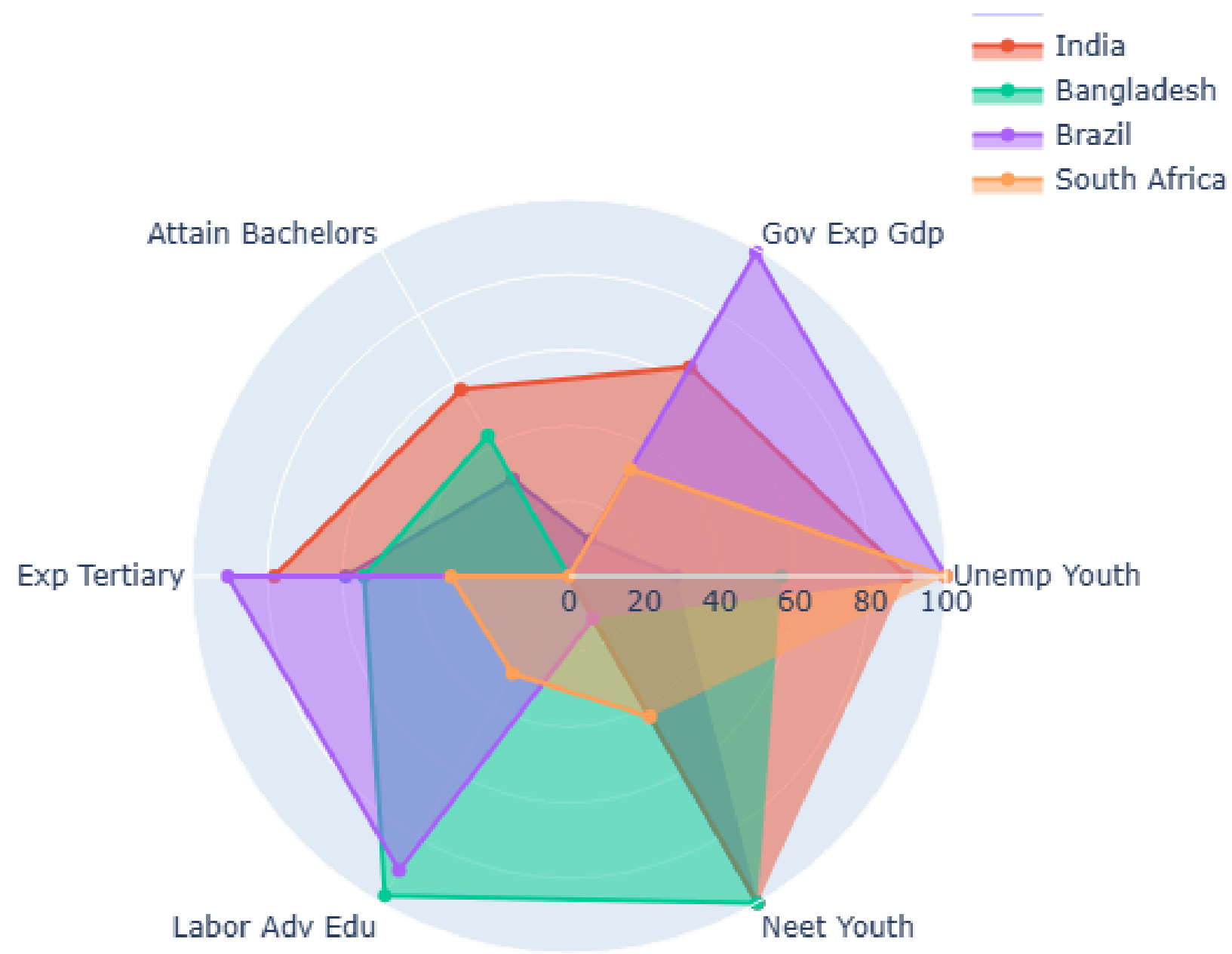


Animated Bubble Chart: Education Spending vs Youth Unemployment Over Time

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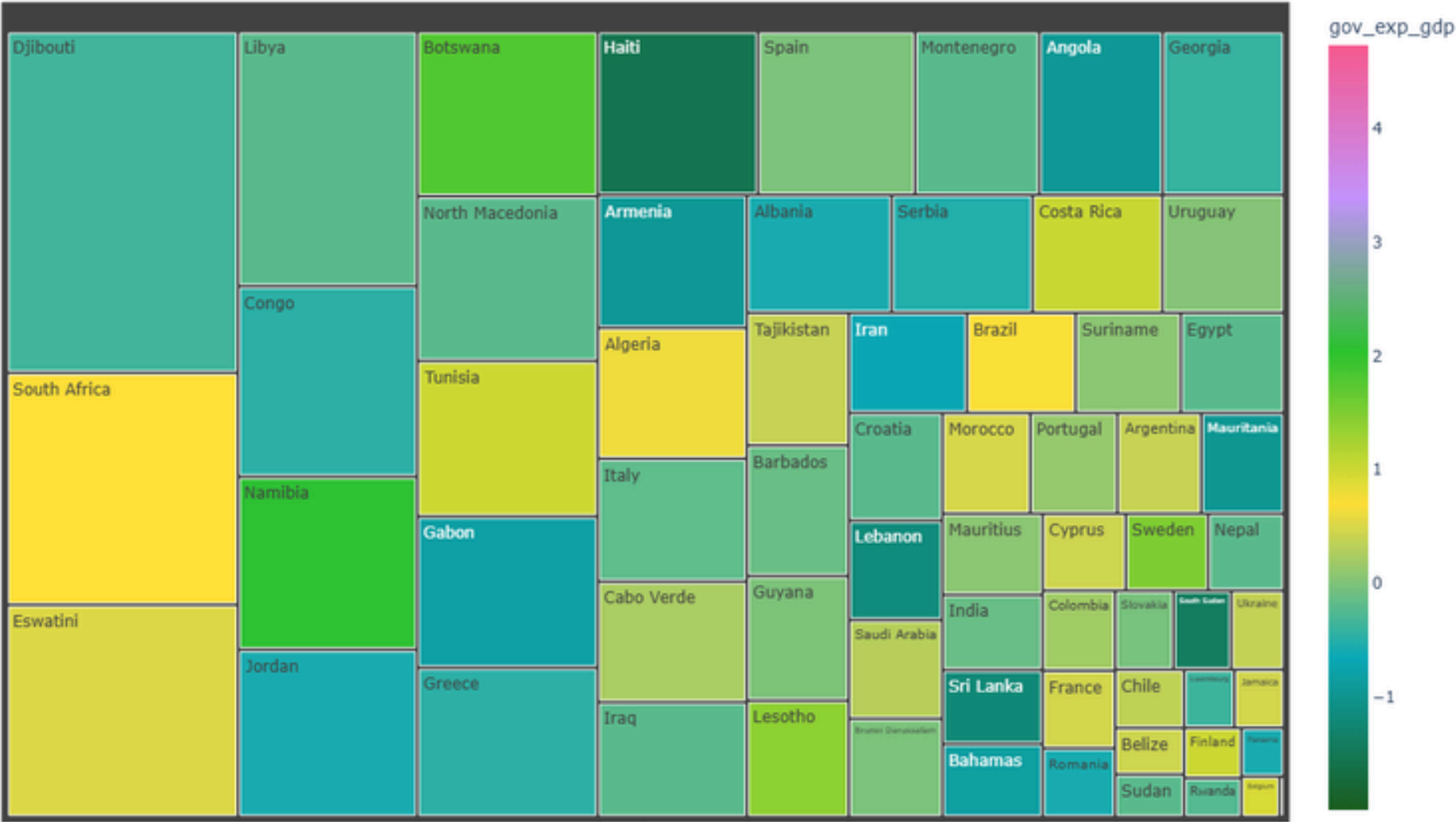


Radar Chart: Multi-Dimensional Country Comparison



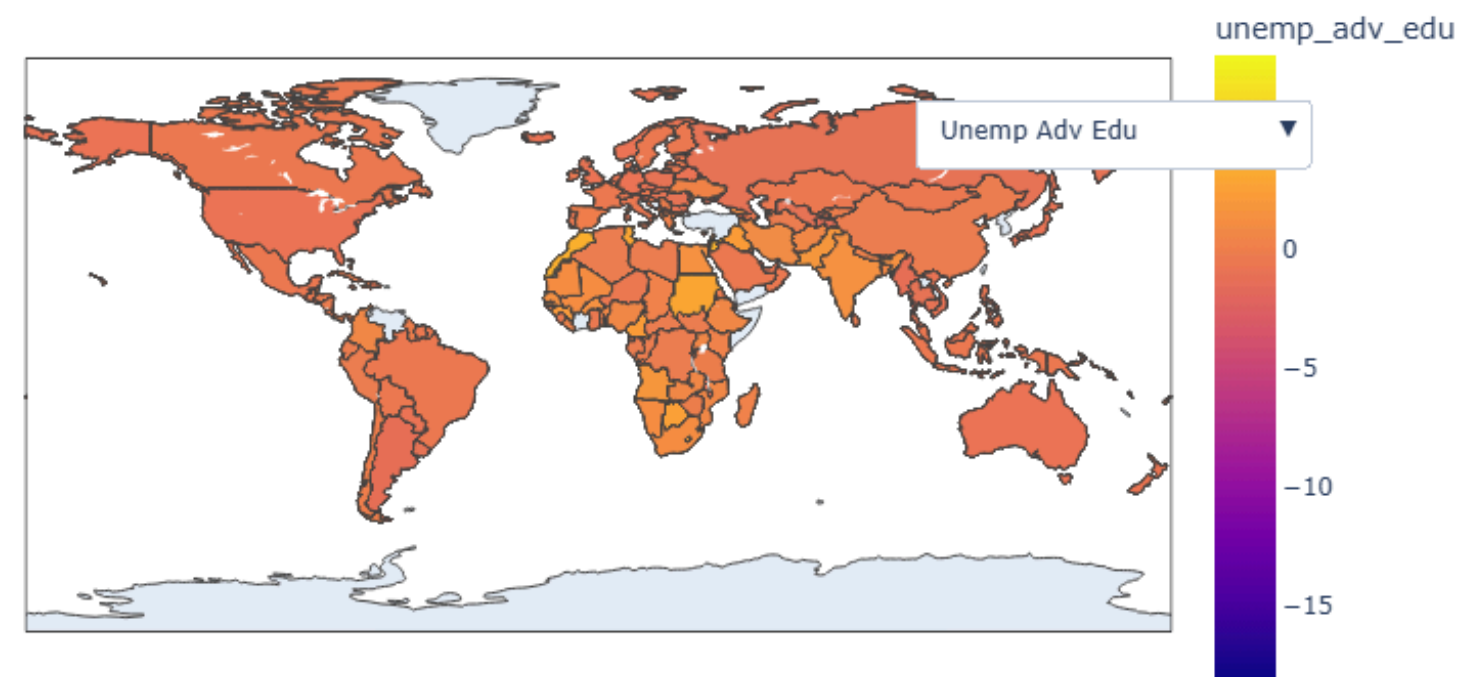
Treemap: Countries by Youth Unemployment (Size) and Gov Spending (Color)

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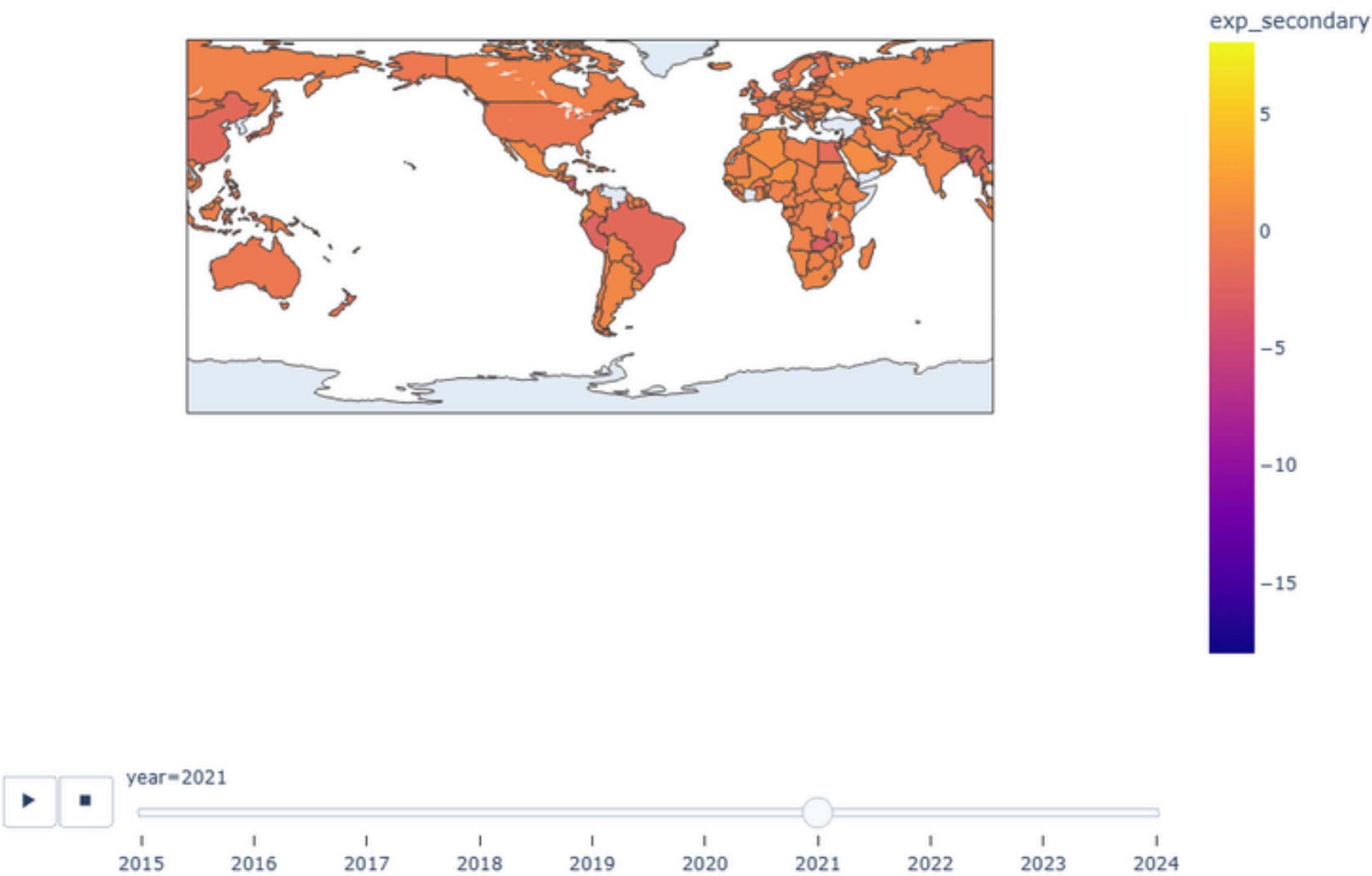


Chroropleth Maps

Global Map - Unemp Adv Edu



Global Map for Exp Secondary Over Time



Policy Recommendations



01

Immediate Actions

Job Creation Focus Prioritize employment generation in sectors capable of absorbing educated youth. Shift from education expansion alone to demand-side labor market interventions.

02

Education Reform

Quality Over Quantity Transform education systems to align curricula with labor market demands. Emphasize vocational training, technical skills, and industry partnerships over enrollment expansion.

03

Regional Cooperation

Coordinated Strategies African nations must develop joint youth employment frameworks. Share best practices, pool resources, and address structural regional challenges collectively.

04

NEET Prevention

Early Intervention Programs Implement targeted initiatives to prevent youth disengagement. Create pathways for unemployed youth to re-enter education, training, or entrepreneurship programs



Conclusion

01

Critical for Development

Youth unemployment threatens economic growth, social stability, innovation

02

Evidence-Based Action Needed

- i. Comprehensive strategies beyond education expansion
- ii. Job creation must match educational output
- iii. Regional and national coordination essential

03

The Path Forward

Transform education systems while simultaneously creating meaningful employment opportunities

THANK YOU

